

APSC 2113 - Engineering Analysis I
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RQ2 - Fall 2026

Solutions and Explanations

General Concepts

For a second-order linear homogeneous ODE of the form

$$ay'' + by' + cy = 0,$$

assume a solution of the form

$$y = e^{sx}.$$

Then

$$y' = se^{sx}, \quad y'' = s^2e^{sx}.$$

Substitution gives

$$as^2e^{sx} + bse^{sx} + ce^{sx} = 0.$$

Since $e^{sx} \neq 0$,

$$\boxed{as^2 + bs + c = 0},$$

which is the characteristic equation.

For a general spring-mass-damper system,

$$my'' + cy' + ky = 0,$$

where

$$m = \text{mass}, \quad c = \text{damping coefficient}, \quad k = \text{spring stiffness}.$$

Dividing by m gives

$$y'' + \frac{c}{m}y' + \frac{k}{m}y = 0.$$

Therefore,

$$\boxed{\text{Damping coefficient per unit mass} = \frac{c}{m}}$$

and

$$\boxed{\text{Stiffness constant per unit mass} = \frac{k}{m}}.$$

The characteristic equation is

$$s^2 + \frac{c}{m}s + \frac{k}{m} = 0.$$

The discriminant is

$$\Delta = b^2 - 4ac.$$

For an overdamped system,

$$\boxed{\Delta > 0},$$

which produces two distinct real roots.

Version 1

Q1

Given

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 5y = 0,$$

compare with

$$ay'' + by' + cy = 0.$$

Thus,

$$a = 1, \quad b = 6, \quad c = 5.$$

Therefore, the characteristic equation is

$$\boxed{s^2 + 6s + 5 = 0}.$$

The coefficients are

$$\boxed{a = 1, \quad b = 6, \quad c = 5}.$$

Q2

For the general spring-mass-damper system

$$my'' + cy' + ky = 0,$$

divide through by m :

$$y'' + \frac{c}{m}y' + \frac{k}{m}y = 0.$$

Therefore,

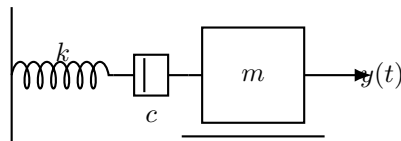
$$\boxed{\text{Damping coefficient per unit mass} = \frac{c}{m} = 6}$$

and

$$\boxed{\text{Stiffness constant per unit mass} = \frac{k}{m} = 5}.$$

Q3

A representative spring-mass-damper system is shown below.



The wall is fixed, the spring has stiffness k , the damper has damping coefficient c , and the block has mass m .

Q4

From Q1,

$$a = 1, \quad b = 6, \quad c = 5.$$

The discriminant is

$$\Delta = b^2 - 4ac.$$

Therefore,

$$\Delta = (6)^2 - 4(1)(5) = 36 - 20 = 16.$$

Since

$$\boxed{16 > 0},$$

the characteristic equation has two distinct real roots.

Hence the system is

$$\boxed{\text{overdamped}}.$$

Q5

The characteristic equation is

$$s^2 + 6s + 5 = 0.$$

Factor:

$$(s + 1)(s + 5) = 0.$$

Thus,

$$s_1 = -1, \quad s_2 = -5.$$

For two distinct real roots, the general solution is

$$y(x) = C_1 e^{s_1 x} + C_2 e^{s_2 x}.$$

Therefore,

$$\boxed{y(x) = C_1 e^{-x} + C_2 e^{-5x}}.$$

Version 2

Q1

Given

$$y'' + 7y' + 10y = 0,$$

we identify

$$a = 1, \quad b = 7, \quad c = 10.$$

Thus,

$$\boxed{s^2 + 7s + 10 = 0}.$$

Q2

For

$$my'' + cy' + ky = 0,$$

division by m gives

$$y'' + \frac{c}{m}y' + \frac{k}{m}y = 0.$$

Hence,

$$\boxed{\frac{c}{m} = 7}$$

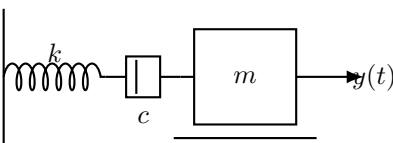
is the damping coefficient per unit mass, and

$$\boxed{\frac{k}{m} = 10}$$

is the stiffness constant per unit mass.

Q3

A representative spring–mass–damper system is



Q4

Here,

$$a = 1, \quad b = 7, \quad c = 10.$$

Calculate the discriminant:

$$\begin{aligned} \Delta &= b^2 - 4ac \\ &= (7)^2 - 4(1)(10) \\ &= 49 - 40 = 9. \end{aligned}$$

Since

$$\boxed{9 > 0},$$

the system is overdamped.

Overdamped

Q5

The characteristic equation is

$$s^2 + 7s + 10 = 0.$$

Factor:

$$(s + 5)(s + 2) = 0.$$

Thus,

$$s_1 = -5, \quad s_2 = -2.$$

Therefore,

$$\boxed{y(x) = C_1 e^{-5x} + C_2 e^{-2x}}.$$

Version 3

Q1

Given

$$y'' + 5y' + 4y = 0,$$

we have

$$a = 1, \quad b = 5, \quad c = 4.$$

Therefore,

$$\boxed{s^2 + 5s + 4 = 0}.$$

Q2

For the general spring-mass-damper system,

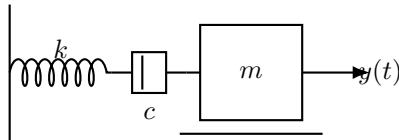
$$\boxed{\text{Damping coefficient per unit mass} = \frac{c}{m} = 5}$$

and

$$\boxed{\text{Stiffness constant per unit mass} = \frac{k}{m} = 4}.$$

Q3

A representative system is



Q4

Using

$$a = 1, \quad b = 5, \quad c = 4,$$

the discriminant is

$$\begin{aligned} \Delta &= b^2 - 4ac \\ &= (5)^2 - 4(1)(4) \\ &= 25 - 16 = 9. \end{aligned}$$

Since

$$\boxed{9 > 0},$$

the characteristic equation has two distinct real roots, so the system is overdamped.

Q5

The characteristic equation is

$$s^2 + 5s + 4 = 0.$$

Factor:

$$(s + 1)(s + 4) = 0.$$

Thus,

$$s_1 = -1, \quad s_2 = -4.$$

Therefore,

$$\boxed{y(x) = C_1 e^{-x} + C_2 e^{-4x}}.$$

Version 4

Q1

Given

$$y'' + 8y' + 12y = 0,$$

we have

$$a = 1, \quad b = 8, \quad c = 12.$$

The characteristic equation is

$$\boxed{s^2 + 8s + 12 = 0}.$$

Q2

For the general spring-mass-damper equation,

$$my'' + cy' + ky = 0,$$

division by m gives

$$y'' + \frac{c}{m}y' + \frac{k}{m}y = 0.$$

Thus,

$$\boxed{\frac{c}{m} = 8}$$

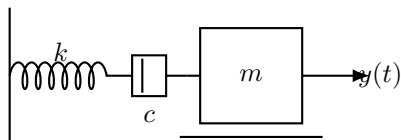
is the damping coefficient per unit mass and

$$\boxed{\frac{k}{m} = 12}$$

is the stiffness constant per unit mass.

Q3

A representative spring-mass-damper system is



Q4

The coefficients are

$$a = 1, \quad b = 8, \quad c = 12.$$

The discriminant is

$$\begin{aligned} \Delta &= (8)^2 - 4(1)(12) \\ &= 64 - 48 = 16. \end{aligned}$$

Therefore,

$$\boxed{\Delta = 16 > 0}.$$

The system is therefore

overdamped.

Q5

The characteristic equation is

$$s^2 + 8s + 12 = 0.$$

Factor:

$$(s + 2)(s + 6) = 0.$$

Hence,

$$s_1 = -2, \quad s_2 = -6.$$

The general solution is

$$y(x) = C_1 e^{-2x} + C_2 e^{-6x}.$$

Version 5

Q1

Given

$$y'' + 9y' + 8y = 0,$$

we identify

$$a = 1, \quad b = 9, \quad c = 8.$$

Therefore, the characteristic equation is

$$\boxed{s^2 + 9s + 8 = 0}.$$

Q2

For a general spring–mass–damper system,

$$my'' + cy' + ky = 0.$$

After dividing by m ,

$$y'' + \frac{c}{m}y' + \frac{k}{m}y = 0.$$

Thus,

$$\boxed{\frac{c}{m} = 9}$$

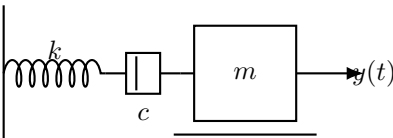
is the damping coefficient per unit mass, and

$$\boxed{\frac{k}{m} = 8}$$

is the stiffness constant per unit mass.

Q3

A representative spring–mass–damper system is



Q4

For Q1,

$$a = 1, \quad b = 9, \quad c = 8.$$

Calculate

$$\begin{aligned} \Delta &= b^2 - 4ac \\ &= (9)^2 - 4(1)(8) \\ &= 81 - 32 = 49. \end{aligned}$$

Since

$$\boxed{49 > 0},$$

the system is overdamped.

Q5

The characteristic equation is

$$s^2 + 9s + 8 = 0.$$

Factor:

$$(s + 1)(s + 8) = 0.$$

Therefore,

$$s_1 = -1, \quad s_2 = -8.$$

Hence,

$$\boxed{y(x) = C_1 e^{-x} + C_2 e^{-8x}}.$$

Version 6

Q1

Given

$$y'' + 10y' + 21y = 0,$$

we have

$$a = 1, \quad b = 10, \quad c = 21.$$

Therefore,

$$\boxed{s^2 + 10s + 21 = 0}.$$

Q2

For

$$my'' + cy' + ky = 0,$$

division by m gives

$$y'' + \frac{c}{m}y' + \frac{k}{m}y = 0.$$

Therefore,

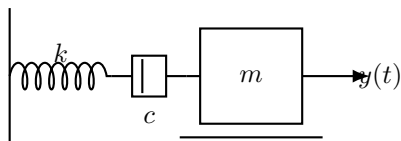
$$\boxed{\text{Damping coefficient per unit mass} = \frac{c}{m} = 10}$$

and

$$\boxed{\text{Stiffness constant per unit mass} = \frac{k}{m} = 21}.$$

Q3

A representative spring-mass-damper system is



Q4

For Q1,

$$a = 1, \quad b = 10, \quad c = 21.$$

The discriminant is

$$\begin{aligned} \Delta &= b^2 - 4ac \\ &= (10)^2 - 4(1)(21) \\ &= 100 - 84 = 16. \end{aligned}$$

Since

$$\boxed{16 > 0},$$

the characteristic equation has two distinct real roots. Therefore, the system is

$$\boxed{\text{overdamped}}.$$

Q5

The characteristic equation is

$$s^2 + 10s + 21 = 0.$$

Factor:

$$(s + 3)(s + 7) = 0.$$

Thus,

$$s_1 = -3, \quad s_2 = -7.$$

Therefore, the general solution is

$$\boxed{y(x) = C_1 e^{-3x} + C_2 e^{-7x}}.$$

Summary Answer Key

Version	Q1	Discriminant	Roots	General Solution
1	$s^2 + 6s + 5 = 0$	16	$-1, -5$	$C_1 e^{-x} + C_2 e^{-5x}$
2	$s^2 + 7s + 10 = 0$	9	$-5, -2$	$C_1 e^{-5x} + C_2 e^{-2x}$
3	$s^2 + 5s + 4 = 0$	9	$-1, -4$	$C_1 e^{-x} + C_2 e^{-4x}$
4	$s^2 + 8s + 12 = 0$	16	$-2, -6$	$C_1 e^{-2x} + C_2 e^{-6x}$
5	$s^2 + 9s + 8 = 0$	49	$-1, -8$	$C_1 e^{-x} + C_2 e^{-8x}$
6	$s^2 + 10s + 21 = 0$	16	$-3, -7$	$C_1 e^{-3x} + C_2 e^{-7x}$

For every version, the discriminant is positive, so every system is overdamped.